Data Science and Artificial Intelligence

Artificial Intelligence

Adversarial Search

Lecture No. 04

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Recap of Previous Lecture

Topic — Informed Search

Topic — (Adversarial Search)

Topic

Topic

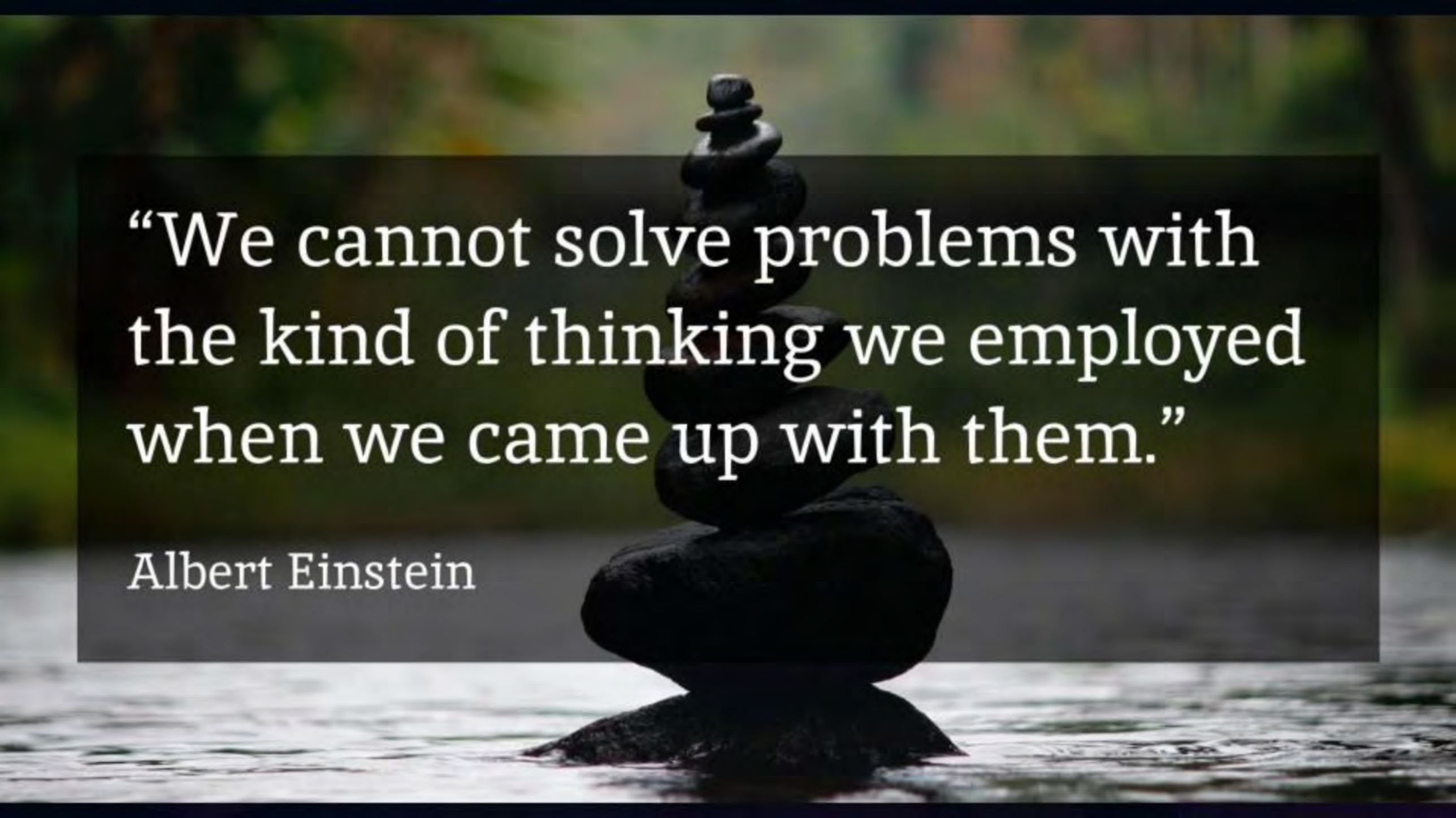
Topic

Turn on Slide map

Topics to be Covered

- Topic
- Topic
- Topic
- Topic
- Topic

Logic ⇒ Propositional


“We cannot solve problems with the kind of thinking we employed when we came up with them.”
Albert Einstein

Proposition logic

- Definition of Propositional Logic : Propositional logic is the branch of logic that deals with propositions, which are statements that are either true or false.
- A proposition is a declarative sentence that is either true or false but not both.
  - Examples : "The sky is blue." (True)
  - "2 + 2 = 5." (False)

⇒ These are statements which are Clearly T/F

- It may Rain today → Can be T/F

⇒ not Propositional logic

Proposition logic

- Q It may rain today. _____ is this proposition logic?

⇒ (we donot know it is T/F)

No

Proposition logic

Simple

- Law of excluded middle (beech main kuch nai ho sakta): If a proposition is not true then it will be false, and similarly if a proposition is not false then it will be true.
- Law of contradiction: A proposition can not be true as well as false simultaneously.

2 type of Proposition ⇒ ① Atomic ② Compound

Proposition logic

- An atomic proposition is a basic statement in propositional logic that cannot be broken down into simpler components.
- These are elementary or simple proposition.
- It is a declarative sentence that is either true or false, but not both.
- In contrast to compound propositions, which are formed using logical connectives (like AND, OR, NOT), an atomic proposition is indivisible.
- For example:
  - "The sky is blue." (Atomic proposition)
  - "It is raining." (Atomic proposition)

ex. The sky is blue and 2+2=5.

Atomic Prop. Atomic Prop.

Combined

$$\underbrace{\text{The sky is blue}}_{\text{atomic prop.}} \ \underline{\text{and}} \ \underbrace{\text{It is Raining}}_{\text{atomic prop}}$$

Proposition logic

- A compound proposition is formed by combining two or more atomic propositions using logical connectives such as AND (∧), OR (∨), NOT (¬), IMPLIES (→), etc.
- Compound propositions are also known as propositional formula pr propositional function
- Let:
  - P: "It is raining."
  - Q: "I will carry an umbrella."

⇒ (If it is Raining then I will carry an umbrella)

⇒ $(P \rightarrow Q)$ implies

⇒ agar P hogo then Q to hoga hi° means $P \rightarrow Q$.

Proposition logic

- A compound proposition could be:
  - $P \wedge Q$: "It is raining AND I will carry an umbrella."
  - $P \vee Q$: "It is raining OR I will carry an umbrella."
  - $\neg P$: "It is NOT raining."
  - $P \rightarrow Q$: "If it is raining, then I will carry an umbrella."
- In each case, the compound proposition connects atomic propositions to create more complex statements.

Proposition logic : Connectives

- Logical Connectives ⇒
- AND (Conjunction): $A \wedge B$ is true if both A and B are true.
- OR (Disjunction): $A \vee B$ is true if at least one of A or B is true.
- NOT (Negation): $\neg A$ is true if A is false.
- IMPLICATION (Conditional): $A \rightarrow B$ is false if A is true and B is false, otherwise true.
- BICONDITIONAL (Equivalence): $A \leftrightarrow B$ is true if both A and B have the same truth values.

$A \wedge B$
$A \vee B$
$A \rightarrow B$
$\neg A$

$A \leftrightarrow B$.

Negation (NOT) ¬

- If p is any proposition, then negation of p/not(p)/ ¬ p is also a proposition whose truth value is true only when p is false and it is false only when p is true,

P ¬P negation of p

1. • P is prop. logic then ¬P is also prop. logic

2.

P	¬P
T	F
F	T

AND (Conjunction): A ∧ B

- If p and q are any two propositions, then "p AND q"/p ∧ q is also a proposition whose truth value is true only when both p as well as q are true.

So A∧B is a also a prop. logic and it is true only when both A and B are True.

A	B	A ∧ B ⇒ A and B
F	F	F
F	T	F
T	F	F
T	T	T

OR (Disjunction): A V B

- If p and q are any two propositions, then "p OR q" /p V q is also a proposition whose truth value is false only when both p as well as q are false.

⇒ So if A, B are logical prop. then A+B, A∨B is also logical prop which is F only if both A, B are false

A	B	A ∨ B
F	F	F
T	F	T
F	T	T
T	T	T

Implication (A → B)

- If p and q are any two propositions, then p implies q (i.e. p → q /if p then q) is also a proposition whose truth value is false only when p is true, and q is false.
- This also means that if P then Q, that is if P is true then Q has to be true then only P implies

A → B ⇒ • If A is true then B has to be true

A	B	A→B
T.	T.	T

Implication (A → B)

$A \to B \Rightarrow$ If It is Raining then I will bring my umbrella

A: It is Raining

B: I will bring my umbrella

A	B	A→B
T	T	T
T	F	F
F	T	T
F	F	T

• If A is false then A→B is "T" bcoz in such cases A→B will not exist.

A → B

Parent → Child

A	B	$\neg A$	$\neg B$	$A \to B$	$\neg A \to \neg B$	$\neg B \to \neg A$	$\neg A \vee B$
T	T	F	F	T	T	T	T
T	F	F	T	F	T	F	F
F	T	T	F	T	F	T	T
F	F	T	T	T	T	T	T

$\neg B \to A \equiv A \to B$

$A \to B \equiv \neg A \vee B$

Biconditional (Equivalence): A ↔ B

- If p and q are any two propositions, then "p↔g" (i.e. p if and only if q) is also a proposition whose truth value is true only when both p and q have same truth value.

- $P \rightarrow q$ if P then Q (Agar P then Q), Q→Child, P→Parent
- $P \leftrightarrow q$ (P if and only if q)

P	Q	P↔Q
T	T	T
F	F	T
T	F	F
F	T	F

Biconditional (Equivalence): A ↔ B

$A \rightarrow B$

A	B	$A \rightarrow B$
T	T	T
T	F	F
F	T	T
F	F	T

$A \Leftrightarrow B$

A	B	$A \Leftrightarrow B$
T	T	T
F	F	T
F	T	F
T	F	F

Tautology

- A tautology is a compound proposition that is always true, regardless of the truth values of its atomic propositions. In other words, no matter what truth values are assigned to the individual propositions, the overall expression will evaluate to true.

Tautology

- Example $P \vee \neg P$

Always True
Tautology.

P	$\neg P$	$P \vee \neg P$
T	F	T
F	T	T

Tautology

- Example $(P \wedge Q) \rightarrow P$ (Yes)

whenever it is F then implies is T

P	Q	$P \wedge Q$	$(P \wedge Q) \rightarrow P$
F	F	F	T
F	T	F	T
T	F	F	T
T	T	T	T

Tautology

- Example $P \rightarrow (Q \rightarrow P)$ ⇒ Tautology.

P	Q	$Q \rightarrow P$	$P \rightarrow (Q \rightarrow P)$
T.	T.	T	T
T	F.	T	T
F.	T.	F	T
F.	F.	T	T

Tautology

- Example $\underbrace{(P \rightarrow Q)}_{A} \vee \neg\underbrace{(P \rightarrow Q)}_{\neg A}$ ⇒ Tautology.

$(A \vee \neg A)$ ⇒ always True

Tautology

- Example $P \vee (P \wedge Q) \leftrightarrow P$

P	Q	$(P \wedge Q)$	$P \vee (P \wedge Q)$	P	$P \vee (P \wedge Q) \leftrightarrow P$
F	F	F	F	F	T
F	T	F	F	F	T
T	F	F	T	T	T
T	T	T	T	T	T

Tautology

- Example $\neg(P \wedge Q) \leftrightarrow (\neg P \vee \neg Q)$

P	Q	$\neg P$	$\neg Q$	$(\neg P \vee \neg Q)$	$\neg(P \wedge Q)$	$\neg(P \wedge Q) \leftrightarrow (\neg P \vee \neg Q)$
F	F	T	T	T	T	T
F	T	T	F	T	T	T
T	F	F	T	T	T	T
T	T	F	F	F	F	T

Tautology

- Example $P \rightarrow (Q \vee \neg Q)$

Tautology ✓

Contradiction

- Contradiction in Propositional Logic
- A contradiction is a compound proposition that is always false, regardless of the truth values of its atomic propositions. In other words, no matter what truth values are assigned to the individual propositions, the overall expression will evaluate to false.

Contradiction

- A contradiction is the opposite of a tautology. It reflects a logical impossibility—something that can never be true. Contradictions are useful in logical reasoning to identify when an argument or a set of premises is inherently flawed or inconsistent.

$(P \wedge (P \rightarrow Q) \wedge \neg Q)$

$(\neg(P \rightarrow Q) \wedge P \wedge Q)$

Contingency in Propositional Logic

- A contingency is a compound proposition that is neither always true (a tautology) nor always false (a contradiction). Its truth value depends on the truth values of its atomic propositions, meaning it can be either true or false based on different assignments of truth values.
- Explanation: Contingent propositions are the most common type of propositions in logic because their truth depends on the specific conditions. Unlike tautologies and contradictions, which have fixed truth values, contingencies vary based on the situation.

Contingency in Propositional Logic

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Contingency in Propositional Logic

- A contradiction is the opposite of a tautology. It reflects a logical impossibility—something that can never be true. Contradictions are useful in logical reasoning to identify when an argument or a set of premises is inherently flawed or inconsistent.
- Identifying a Contingency Truth Table: A proposition is a contingency if its truth table contains both true and false outcomes. This means the truth value depends on specific cases. Logical Status: Contingencies are neither universal truths (like tautologies) nor universal falsehoods (like contradictions).

Contingency in Propositional Logic

- Example 1: $(P \wedge Q)$

Contingency in Propositional Logic

- Example 1: $(P \wedge Q)$

THANK - YOU